



me

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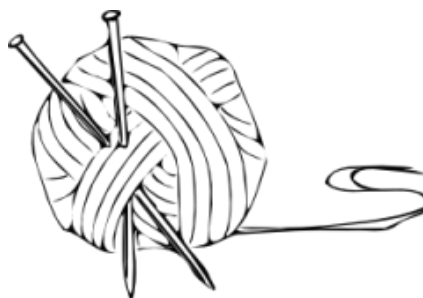
Demos

# Minimal examples

Examples for  
Rnw,

Markdown, HTML and LaTeX

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For beginners, it is probably a good idea to start with some minimal examples; here I provide a few examples for Rnw, LaTeX, Markdown and HTML, respectively.

## How is a report generated from R code

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Regardless of which format you use, the basic idea is the same: **knitr** extracts R code in the input document, evaluates it and writes the results to the output document. There are two types of R code: chunks (code as separate paragraphs) and inline R code. For example, here we show a code chunk (using traditional Sweave syntax):

Example text outside R code here; we know the value of `pi` is `\Sexpr{pi}`.

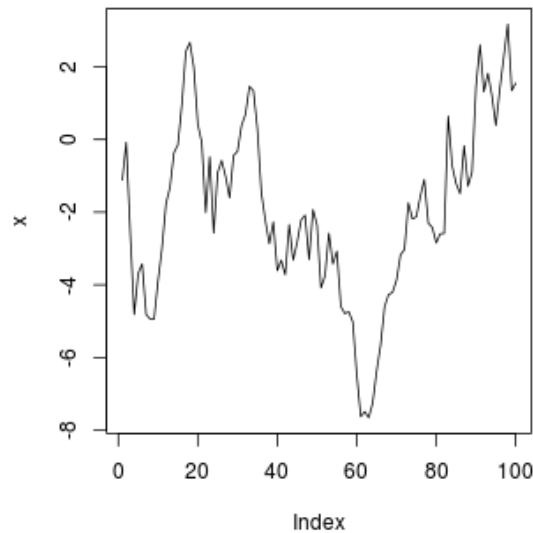
```
<<my-label, eval=TRUE, dev='png'>>=
set.seed(1213) # for reproducibility
x = cumsum(rnorm(100))
mean(x) # mean of x
plot(x, type = 'l') # Brownian motion
@
```

Other text outside R code here.

After weaving this chunk, we get the output as below. Compare them and you will realize the inline code `\Sexpr{pi}` is replaced by the value of `pi`, and the code chunk is also evaluated -- numerical results are printed and plots are inserted in the output as well.

Example text outside R code here; we know the value of `pi` is 3.1415926.

```
set.seed(1213) # for reproducibility
x = cumsum(rnorm(100))
mean(x) # mean of x
## [1] -1.939758
plot(x, type = 'l') # Brownian motion
```



Other text outside R code here.

Now hopefully you have got an idea of how automatic report generation works. There are many options of which we can make use to tune the results, for example, we can use `echo=FALSE` to hide the R source code (usually you do not want R code to appear in a final report, unless you are writing a tutorial on R), or `results='hide'` to hide the printed results (e.g. you will not see `## [1] -1.939758` above if you use this option), or control the format, size and alignment of plots. Beside local chunk options which you write between `<<` and `>>=`, you can also set options globally like:

```
knitr::opts_chunk$set(echo=FALSE, fig.path='myproject/plot-  
' , cache=TRUE)
```

Once options are set globally, all the following chunks will be affected by these options, so if you use an option frequently in many chunks, you may want to set it globally.

The advantage of using **knitr** is obvious: you only maintain the source code, and whenever you want a report, you just knit the source code, and everything will be generated automatically (tables, plots and numbers in lines). There is no need to manually copy and paste anything. Next time if your data source is changed, you simply run the process again, and all results can be updated. Let computers do the tedious

job, because this is what they are good at. Humans should focus on other jobs like the statistical analysis and organization of the report.

## Examples

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Two main differences among different file formats are: they use different syntax to wrap up R code in the input document (see [patterns](#)), and the results returned from R are marked up according to the syntax of the output document (see [hooks](#)). For instance, Rnw uses `<<` and `>>=`, and HTML uses `<--!begin.rcode` and `end.rcode-->` for the input; plots will be put in `\includegraphics{...}` for LaTeX output, and in `<img src=...>` for HTML output.

- Rnw
  - Rnw source: [knitr-minimal.Rnw](#)
  - PDF output: [knitr-minimal.pdf](#)
  - a more minimal [002-minimal.Rnw](#) (recommended for absolute beginners)
- Markdown
  - example 1: [001-minimal.Rmd](#)
  - output: [001-minimal.md](#) (GitHub does the job of parsing the md file to HTML)
  - example 2: [knitr-minimal.Rmd](#)
  - output: [knitr-minimal.md](#)
- HTML
  - html source: [knitr-minimal.Rhtml](#)
  - output: [knitr-minimal.html](#)
- LaTeX
  - tex source: [005-latex.Rtex](#)
- reStructuredText
  - source: [knitr-minimal.Rrst](#)
  - output: [knitr-minimal.rst](#)

You can run `knit()` to knit the input file, e.g.,

```
library(knitr)
knit('knitr-minimal.Rnw')
knit('knitr-minimal.Rhtml')
```

```
knitr('knitr-minimal.Rmd')
```

Note for the HTML demo, you need to install the **rgl** package to get the snapshot in the output.

## Some comments

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One consideration in defining the default syntax for these files is that we should try not to break the syntax of other software systems, e.g. for LaTeX and HTML, the chunk options are written in comments, so the original document is still a valid TeX or HTML document. Unfortunately the original author(s) of literate programming did not seem to care about this issue at all, and invented the `<<>>= + @` syntax; then Sweave added more pseudo-LaTeX commands like `\SweaveOpts` and `\Sexpr`; we cannot do much about them since they have been used for many years, and we are likely to be in bigger troubles if we break these rules. The LaTeX example above only serves as an example of how I would have defined the syntax, and you may not really want to follow it.

As a final side note, the extension `Rnw` is a combination of `R` + `nw` where `nw` stands for `Noweb` (= `No` + `web`), and `WEB` was coined by Donald Knuth to denote the idea of weaving documents. Today probably everybody will think of `WWW` when seeing the word `web`.

If you have any questions, please consider asking them on *StackOverflow*, where you may get more attention and fast answers.

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**hcheng** · 2 years ago

Hi, I am very confused why the results are different when I use markdown and just run in Rstudio. In your markdown example, there are two lines: `set.seed(123) rnorm(5)`. I get quite different results by running markdown and rstudio. For markdown, the results are what you provide in the output but in rstudio the results are 0.1870696 1.9960295 2.2993699 1.1274577 1.0811140. As I need to generate some simulated dataset, If i use the markdown, the results are far too different.

Could you give me any clue?? Thanks!

1 ^ | v • Reply • Share ›



**Yihui Xie** Mod → hcheng • 2 years ago

That is unexpected. Are you sure you also copied `set.seed(123)` into RStudio? In other words, did you run exactly the same code in R Markdown and in RStudio console?

^ | v • Reply • Share ›



**Yuanchun** → Yihui Xie • 8 months ago

Why the compile pdf does not show those plots?

1 ^ | v • Reply • Share ›



**hcheng** → Yihui Xie • 2 years ago

Yes, I am pretty sure. Can you try ? I get the same result using R or Rstudio, but different in markdown. Even for the following lines , the result of Rstudio or R is `[1] 3.612922`. So I'm quite confused.

```
set.seed(1213) # for reproducibility
```

```
x = cumsum(rnorm(100))
```

```
mean(x) # mean of x
```

```
## [1] -1.939758
```

^ | v • Reply • Share ›



**Yihui Xie** Mod → hcheng • 2 years ago

No, I still cannot reproduce it. Can you put `sessionInfo()` in your R Markdown code, and compare that with `sessionInfo()` in your R console?

^ | v • Reply • Share ›



**JenHao Wu** → Yihui Xie • a year ago

Hi Yihui,

I have a similar problem. Instead of having different values from `rnorm()` which I have the same value after setting the same `set.seed()` in the set code chunk section in .Rnw file and in Rstudio, I have different results from WinBUGS (through R2WINBUGS package) even set the same `bugs.seed`. Do you have any suggestion for this problem? Thank you very much.

^ | v • Reply • Share ›



**Yihui Xie** Mod → JenHao Wu • a year ago

Do you mean WinBUGS and R generate different random numbers with the same seed? I do not know why that surprised you. R and WinBUGS should not be using exactly the same RNG's.

^ | v • Reply • Share ›



**JenHao** → Yihui Xie • a year ago

Thank you for your promptly reply, Sorry I didn't make it clear in the beginning. I wrote my R code in Rstudio and link to WinBUGs through R2WinBUGs package, and now I call the same R script in Latex through knitr. And I have different result from running in Rstudio and Latex. I hope this is clear enough to let your know my problem. Thank you so much.

1 ^ | v • Reply • Share ›



**juan** • a month ago

when will it be possible to use ggvis on Rnw knitr documents to produce pdf? (With Windows, where it's not possible to get a working Vega)

^ | v • Reply • Share ›



**ManiA** • 2 months ago

Hi, I have a question which I asked here <http://stackoverflow.com/quest...> do you have any hint for me please?

^ | v • Reply • Share ›



**@γσ, Eng** • 3 months ago

paragraph 1...  
````{r reset, include=FALSE}`  
`gc(reset=TRUE)`  
`````

paragraph 2...

I got a question about recall chunk. May I know how do I recall the chunk `reset` after paragraph 2?

^ | v • Reply • Share ›



**Young** • 6 months ago

**@Yihui Xie** Hi, I am trying to learn knitr. I started with your basic example "knitr-minimal.Rnw"; however, the pdf file I get does not include the figures. I am running exactly what you have included without any modeification. What is the problem?

^ | v • Reply • Share ›



**Yihui Xie** Mod → Young • 6 months ago

You are most likely to be using Sweave instead of knitr: <http://yihui.name/knitr/demo/r...>

^ | v • Reply • Share ›



**Stefano Lombardi** • 8 months ago

Hi Tihui,  
 I have a basic question. I am trying to replicate your 002-minimal.Rnw example.  
 After changing the weaving option to "knitr", I tried to compile the pdf, but I get this:

Quitting from lines 9-13 (knitr\_minimal.Rnw)  
 Error in data.frame(..., check.names = FALSE) :  
 arguments imply differing number of rows: 3, 0

Note that if I delete the chunk of R code the pdf is compiled correctly.

Actually, whenever I try to embed any chunk of R-code the above error is reported.

Do you have any idea why this happens?

Thank you very much for your work!  
 best,

Stefano

^ | v • Reply • Share ›



**Xing** • 10 months ago

Hi, Yihui, Can I just the R markdown Beamer-Presentation in Chinese?

I cannot make it

^ | v • Reply • Share ›



**David States** • a year ago

Would like to use LyX to edit knitr .Rnw files but having some problems getting started. Have R version 3.1.1 installed with the knitr library installed on a Mac running OS X 10.10.2 (Yosemite).

First, LyX will not open .Rnw files directly, only .tex files. I guess this is by design, but it would be very useful if LyX could load .Rnw files directly and autoload the knitr module in the process.

Second issue, if I follow the description on this website, open a new file in LyX, load the knitr module and then try to import an existing .Rnw file (knitr-minimal.Rnw from this website), I get an error message from R:

An error occurred while running:

```
Rscript --verbos --no-save --no-restore
"Applications/LyX.app/Contents/Resources/scr..."
"/Users/dstates/Documents/""knitr-minimal.Rnw"
"/Users/dstates/Documents/""knitr-minimal.tex"
"/Users/dstates/Documents"
```

Sorry if I am missing something really basic here. Newby :(

^ | v • Reply • Share ›



**Paul Rougieux** → David States • 10 months ago

Start with the basic .lyx file knitr.lyx located on this page: <http://yihui.name/knitr/demo/l...>

^ | v • Reply • Share ›



**Honza** • 3 years ago

I completely agree with you on the advantage of the LaTeX example with R codes written in comments. But unfortunately Emacs ESS does not support it. Or does it?

^ | v • Reply • Share ›



**Yihui Xie** Mod → Honza • 3 years ago

No, it does not. The inertia is very strong :)

^ | v • Reply • Share ›



**Marcus R** • 3 years ago

Hi. I am about to write my masters thesis. I am right now setting up my software environment with git, R, latex etc within a Ubuntu 12.04 installation. I used Sweave for my Bachelors thesis but I will now give knitr a try. But I have some problems.

I decided to first try try replicate the oo2-minimal.Rnw above with the commands:

```
Rscript -e "library(knitr); knitr('oo2-minimal.Rnw')"  
pdflatex oo2-minimal.Rnw.tex
```

But then I get a error that says:

! Undefined control sequence.

```
l.59 \SweaveOpts  
{concordance=TRUE}
```

And in the resulting pdf there is this text above that says:  
concordance=TRUE

I know you have a guide for R Studio to output PDF (even though I still have trouble with it). But do you have, or know about, some guide that how to use command line to get a good PDF-output. I know there are pdfLatex, knitr2pdf and a lot of different programs but there is so much to get into.

And also, I almost forgot, thanks a a lot for this package, when I get hold of it it really going to turn my thesis studies from frustration to Euphoria! :)

^ | v • Reply • Share ›



**Yihui Xie** Mod → Marcus R • 3 years ago

Delete that line: `\SweaveOpts{concordance=TRUE}` (it is useless for knitr) Note I do not have that line in my oo2-minimal.Rnw example.

^ | v • Reply • Share ›



**Marcus R** → Yihui Xie • 3 years ago

Thanks a lot, I don't know how it got into the code. Now the simple script runs without problem.

May I ask another question. I intend to write my thesis in such a way that I can generate both Thesis, Presentation and other files out of the same file as described here:  
<http://people.su.se/~lundh/pro...>

Professor Lundholm uses docstrip as a way to produce multiple output. Do you know of any other techniques to do this or is docstrip the way to go?

^ | v • Reply • Share ›





**Yihui Xie** Mod → Marcus R  
• 3 years ago

Article <--> tech report is easy (as in Lundholm's article); in knitr you can use child documents to decide whether to include certain parts of the document:  
<http://yihui.name/knitr/demo/c...>

But I think thesis <--> presentation from the same file is much more challenging because they are very different. However, it is easy to share code chunks between two documents if you use this:  
<http://yihui.name/knitr/demo/e...> I know beamer has some support on conversion between article and presentation, but it does not make much sense to me. I'd rather start writing the presentation from scratch.

^ | v • Reply • Share ›



**Bastiaan Quast** • 3 years ago

Thank you very much for this, I tried to replicate your minimal.Rmd example and upload it to Github, but the pictures won't display. I guess this is because they are in the figure folder, is there a way to avoid this?

^ | v • Reply • Share ›



**Yihui Xie** Mod → Bastiaan Quast • 3 years ago

It is tricky if you do not understand how Github handles the content in the repositories. For images, you have to use their *\*raw\** URL's. For example, the raw link of this image <https://github.com/yihui/knitr...> is



or <https://github.com/yihui/knitr...> You have to let knitr know this change in the URL (the default direct link will not work). For beginners, here is a simple solution: <http://yihui.name/knitr/demo/u...> Once you know Github well enough, you can set the root.url option: <http://yihui.name/knitr/option...> (e.g. <https://github.com/yihui/knitr...>

The other even easier approach is to use RStudio and RPub.com.

^ | v • Reply • Share ›



**AJ Rossini** • 3 years ago

actually, you got it a bit wrong. The original noweb syntax (thanks to Norman Ramsey) was to have code chunks and doc chunks interspersed, with means to have code chunks in other code chunks, but left the doc chunks unlabelled. From this, you could use any doc language (but needed to be consistent, no facility for switching) and any code language (which could

Minimal examples: Examples for Rnw, Markdown, HTML and LaTeX | knitr  
 no facility for switching) and any code language (which could be inconsistent, since there was a facility for switching and annotation. You should read Norman's tech report (and I think he had a publication that is citable) for that. Sweave broke the noweb syntax by mixing up code in the doc chunks, but this wasn't a bad idea in theory, since Fritz wanted to do reporting. If I had started with nuweb (which was a cleaner but trickier code-language agnostic language), we'd have made a bit more progress..

best,  
 -tony

^ | v • Reply • Share ›



**Yihui Xie** Mod → AJ Rossini • 3 years ago

Thanks so much, Tony. I'm excited to see an expert in this area like you here.

I remember I read the two articles on Norman Ramsey's website when I started re-inventing the wheel last year: <http://www.cs.tufts.edu/~nr/no...> (is there another report I missed?) I do not quite understand your comments about noweb, or perhaps I did not make myself clear enough. I do not actually have much complaint about the noweb syntax; I mean I would have used something like % chunk.begin and % chunk.end instead of <<>>= and @, but that is more like a personal taste instead of a really big problem.

What I do not like are \SweaveOpts{} and \SweaveInput{}. They could have been avoided by properly using code chunks (in knitr, I use opts\_chunk\$set() to set global chunk options, and <<child='file.Rnw'>> to include child documents). Pseudo LaTeX commands bring additional parsing difficulties for developers, and require additional memory of users to remember new LaTeX commands (and since they look like LaTeX commands, they can be odd to appear in a literate programming document in HTML or other doc formats).

Thanks for pointing out nuweb. I saw that just a few days ago, and initially I thought it was a typo of noweb :-). I just glanced at its user manual, and it seems to be a huge improvement over noweb. I'm wondering if we

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